Hari Chandra Wijaya

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Education

Feng Chia University — Taichung, Taiwan

B.S. in Informatics Engineering and Computer Science – GPA: 4.03/4.30 — *Expected Jun 2027*

- Ranked 3rd of 54 in class. Coursework: Algorithms, Databases, Systems Programming, Computer Architecture, Microprocessors

Projects

Arus | *React, TypeScript, FastAPI, Python, PostgreSQL, NumPy, Docker* — Jul 2026 – Present

- Wrote the full codebase of a two-person portfolio tracker covering the ∼950-stock IDX (Indonesian exchange) universe, with time-weighted returns, risk analytics, and IHSG benchmarking
- Implemented Markowitz mean-variance optimization from scratch via projected gradient descent with exact simplex projection, no heavyweight solver; efficient-frontier math co-designed with an actuarial-science collaborator
- Modeled expected returns with CAPM or annualized log returns, solving minimum-variance, target-return, and maximum-Sharpe portfolios from one parameterized frontier
- Deployed on a self-managed Linux VM with Docker Compose and Caddy (automated TLS); tickers lazily backfill 5 years of history on first use, then sync nightly
- Designed transactions as the single source of truth, deriving holdings and cash balances in SQL; covered by 200+ backend tests (source)

Aegis | *React, TypeScript, Tailwind CSS, FastAPI, Python, OpenCV, NumPy* — Apr 2026 – Present

- Senior capstone: full-stack media authentication platform embedding invisible watermarks in images and video
- Engineered an LWT + DCT + SVD + QIM watermarking pipeline with BCH(15,7) error correction, imperceptible at 41.9 dB mean PSNR and 0.993 SSIM
- Localized tampering with per-block SHA-256 parity, catching 29 of 30 content edits in a 190-case evaluation, and swept detection thresholds against false alarms from compression, noise, and resizing
- Tagged each video frame with an ID and chain bits, exposing dropped or reordered frames (source)

WasteWise | *Next.js, TypeScript, FastAPI, Python* — Jul 2026

- One of two developers on a 5-person team; built in 5 days for AMD Developer Hackathon: ACT II (Unicorn Track)
- Built the Next.js frontend of an AI restaurant purchasing assistant: setup wizard with CSV upload and Google Maps location picker, demand-forecast dashboard, and backtest replay chart
- Replaced a deprecated USDA price source with a FRED retail benchmark, added currency conversion for non-USD price data, and tuned the LLM demand-adjustment agent (source)

Finance-Buddy | *Java, Kotlin, Android, Firebase, Firestore, OpenAI, Google Places* — Spring 2026

- Built an Android finance app on a Firestore data model; a GPT-4o-mini assistant answers questions over the user's own data and Google Places suggests food within budget (source)

Experience

Student Assistant | *Information Engineering Dept., Feng Chia University* — Sep 2024 – Present

- Handle inter-unit official correspondence and meeting logistics for the department office, working in Mandarin

Honors & Awards

Outstanding Student Scholarship, Feng Chia University — *1st in class in 2 of 3 semesters* — Apr 2025 – May 2026

Outstanding Overseas Chinese Student Scholarship, Overseas Community Affairs Council, Taiwan — Nov 2025

Overseas Chinese Student Grant, Feng Chia University — *3 years; top tier NT$120,000/yr* — 2024 – 2027

Skills

Programming Languages: Python, TypeScript, JavaScript, SQL, Java, Kotlin, Bash
Frameworks & Libraries: React, Next.js, FastAPI, SQLAlchemy, NumPy, OpenCV, pytest, Android SDK
Coursework: C, C++, Assembly, PyTorch, TensorFlow/Keras, scikit-learn
Tools & Platforms: Git, Docker Compose, Caddy, Linux, PostgreSQL, SQLite, Firebase, Vercel, Render, Oracle Cloud
Spoken Languages: Indonesian (native), English (TOEIC 990/990), Mandarin Chinese (professional)